

Order Araneae continued

Family Salticidae

Jumping spiders

Occurrence 5,500 spp. worldwide; in grassland, meadows, woodland, forests, gardens



The 5,000 species of jumping spiders (salticids) make up one of the largest of the spider families. Most jumping spiders have stout, hairy bodies, 1/16–1/2 in (2–16 mm) long, and are often drab in appearance, although many tropical species are vividly colored, with elaborate markings. Some species are very antlike, mimicking their prey in both looks and behavior. Although the different species do vary greatly, a distinctive feature common to all is their prominent eyes. At the front of the carapace, which is rectangular, with a rounded rear and square front, sits a row of 4 large eyes. The middle 2 eyes, which are very much larger than the others, resemble old-fashioned car headlights. These eyes enable jumping spiders to judge



EUOPHRYS SPECIES

Members of the genus Euophrys are often found on the ground, among grass and low-growing plants, or under stones. As in many species, males fight for access to a female, the larger male usually winning.

JUMPING

Salticids jump not only to catch prey but also to escape from predators. The hind pair or 2 hind pairs of legs are extended rapidly by hydraulic pressure.



ATTACKING PREY

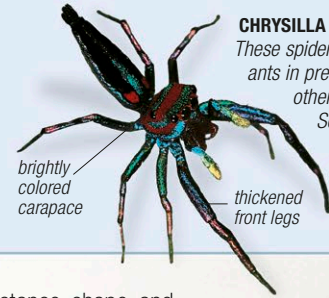
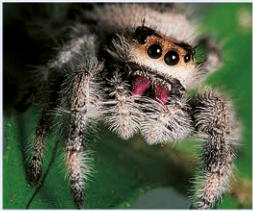
Before it leaps, a jumping spider attaches a safety line of silk in case it gets carried off course. This species is known as the zebra spider on account of its striped markings. The mouthparts (chelicerae) of males are larger than those of the females.

STRIKING APPEARANCE

Some tropical jumping spiders are very brightly colored, with iridescent hairs arranged in patches and spots. In some, the whole of the upper surface of the abdomen may be bright red or orange. Males often have enlarged front legs and ornamented pedipalps, which they use to attract the attention of females.

PHIDIPPUS REGIUS

Males and females of this species, native to S.E. North America, communicate using more than 20 different signals made with their legs and palps. Other spiders in the genus do the same.



CHRYSILLA SPECIES

These spiders catch and eat ants in preference to any other kind of insect. Some species jump on their prey from behind, while others tackle the ant head first.

distance, shape, and movement very accurately, and give them the most keen vision of any spider. The back of the eye capsule can be moved inside the head to keep the image of the prey centered on the retina. The smaller outside eyes and a pair toward the rear of the carapace simply detect movement. Jumping spiders are active during the day, stalking insect prey on the ground, on walls, and in bushes, especially in warm, sunny locations. Their characteristic hunting method of jumping on their prey to seize it gives them their common name. When the weather is bad, they retreat to a small, silken nest they make in a crevice or crack. The spiders use similar shelters over winter and when they molt or lay eggs. The eggs, laid among vegetation, bark, and moss, are wrapped in silk, and the female guards them until they hatch.



BAGHEERA KIPLINGI

This jumping spider is the only known herbivorous spider. In its native Central American forests, Bagheera kiplingi lives on acacia trees that produce nutrient-rich nodules to sustain colonies of ants. In turn, the ants clear the acacia of leaf-eating insects. The spider feeds from the nodules, too—but without providing the tree with anything in return.

BRIGHT PATTERN

Maratus bubeo is one of several species of jumping spiders known as peacock spiders, which are found in western Australia. The bright coloration on the abdomen of the male peacock spider is structural color, and not due to pigments.

